

ArduPilot MAGFit in flight compass calibration

This tool takes a .bin log and performs a "MAGFit inflight calibration". Calibration parameters are found that fit the measured magnetic field from the log to the expected field from the World Magnetic Model. For best results use a flight log that covers as many orientations as possible. Flying figure of eight patterns works well. If you need a motor calibration using either throttle or a current reading then you should also cover as wide a range of throttles and current draws as possible. You do not have to be in any particular flight mode.

Setup

Analysis time ?

Start (s) 366.35960

End (s) 1155.5297

Attitude source ?

☐ DCM
 ☒ EKF 3 IMU 1

Choose File

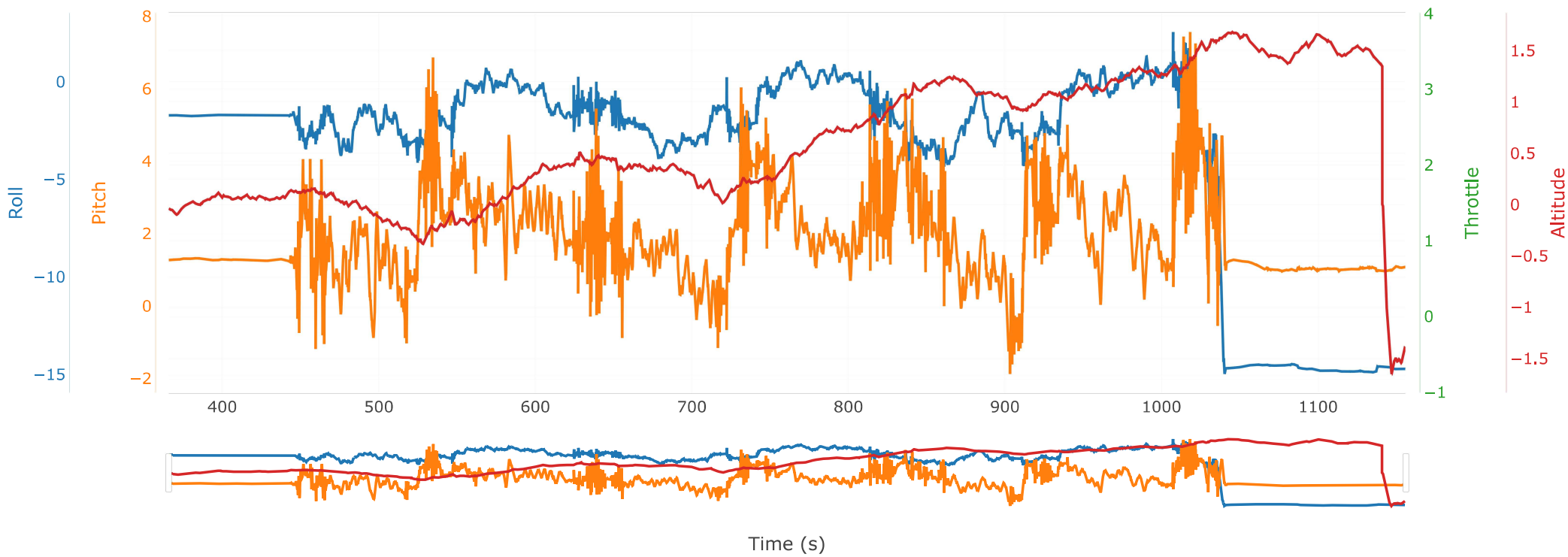
2026-07-06 15-38-17.bin

Open In

Calculate

?

Flight Data ?



Expected vs measured body frame magnetic field ?

Compass 1

QMC5883L via I2C

Use: ☒, External: ☒, Health: ☒

Coverage:

Parameter changes ?

Use sensor: ☒ No change, ☐ Use, ☐ Don't use

Orientation: ☒ Check, ☐ Fix 90, ☐ Fix 45

Calibrations ?

☒ Existing

No motor comp ?

☒ Offsets

☐ Offsets and scale

☐ Offsets and iron

Compass 2

IST8310 via I2C

Use: ☒, External: ☒, Health: ☒

Coverage:

Parameter changes ?

Use sensor: ☒ No change, ☐ Use, ☐ Don't use

Orientation: ☒ Check, ☐ Fix 90, ☐ Fix 45

Calibrations ?

☒ Existing

No motor comp ?

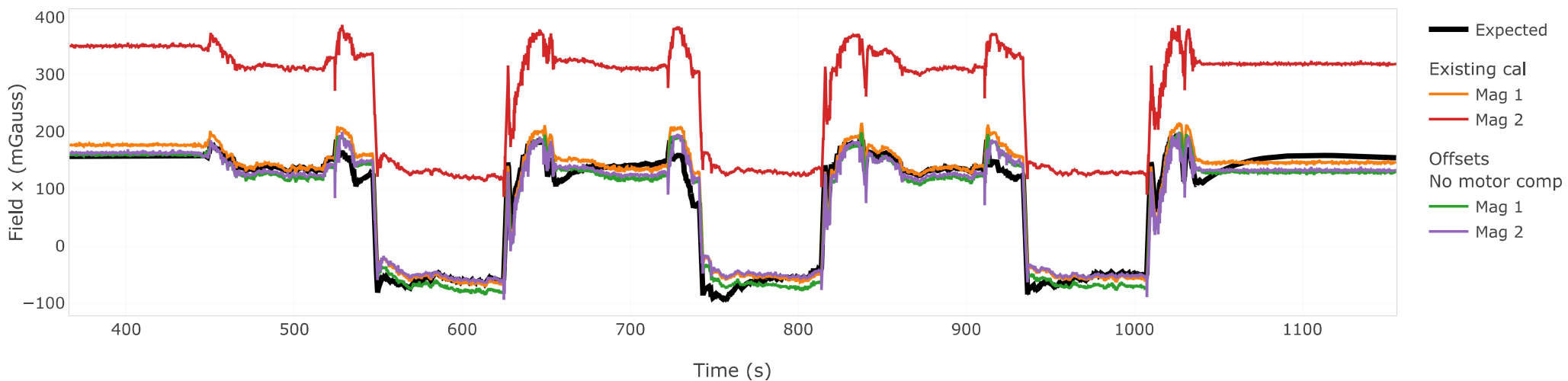
☒ Offsets

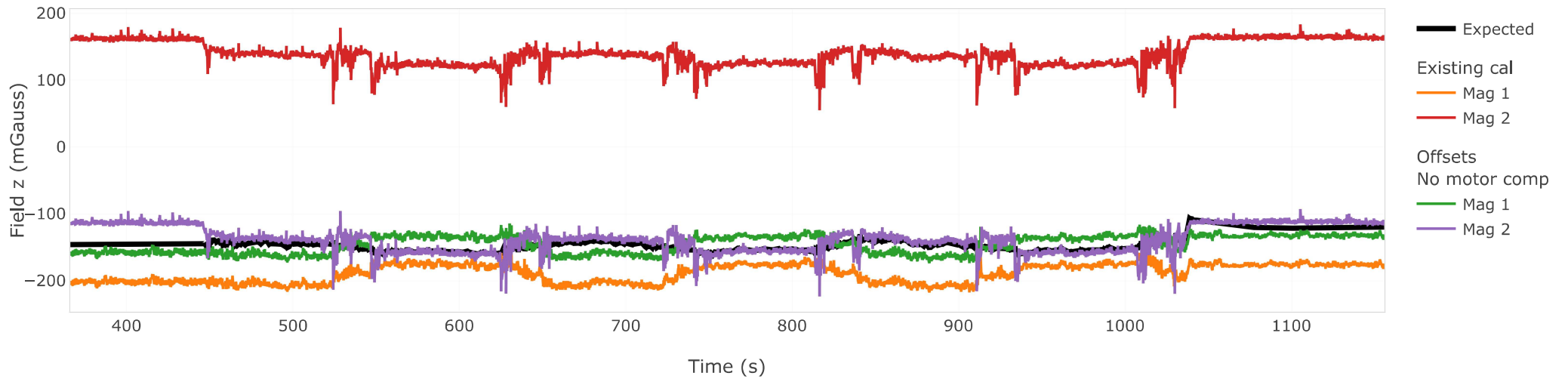
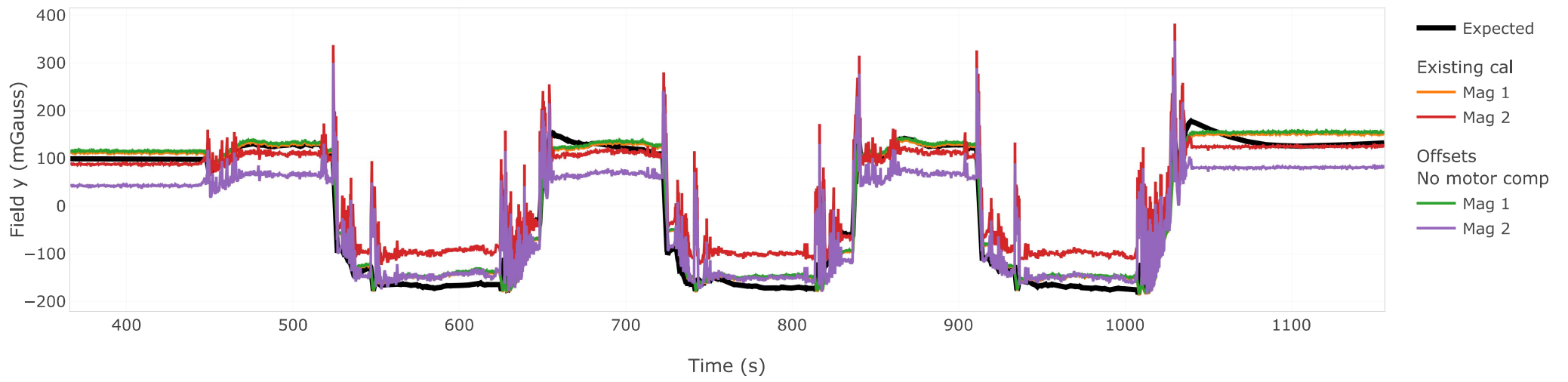
☐ Offsets and scale

☐ Offsets and iron

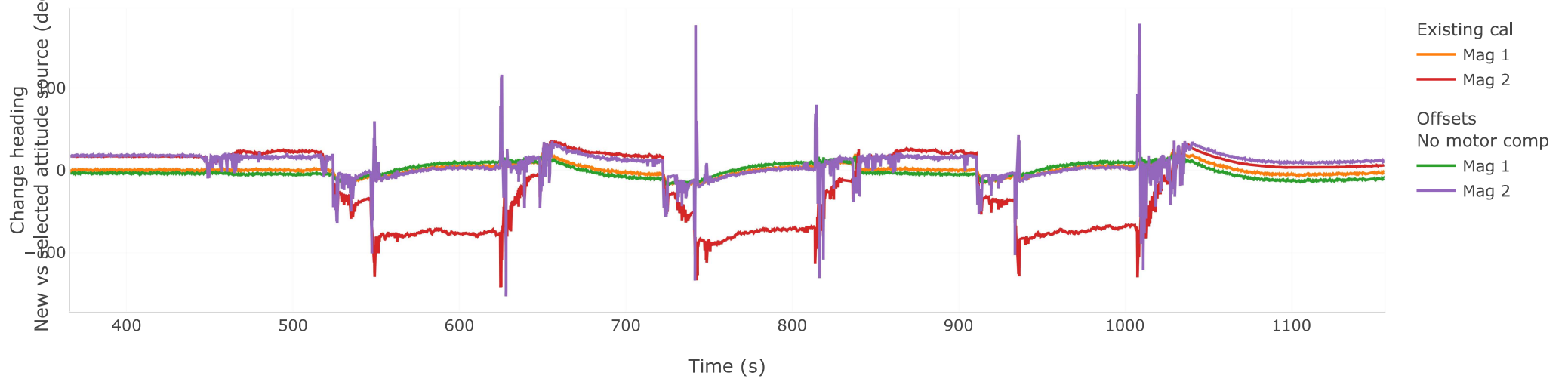
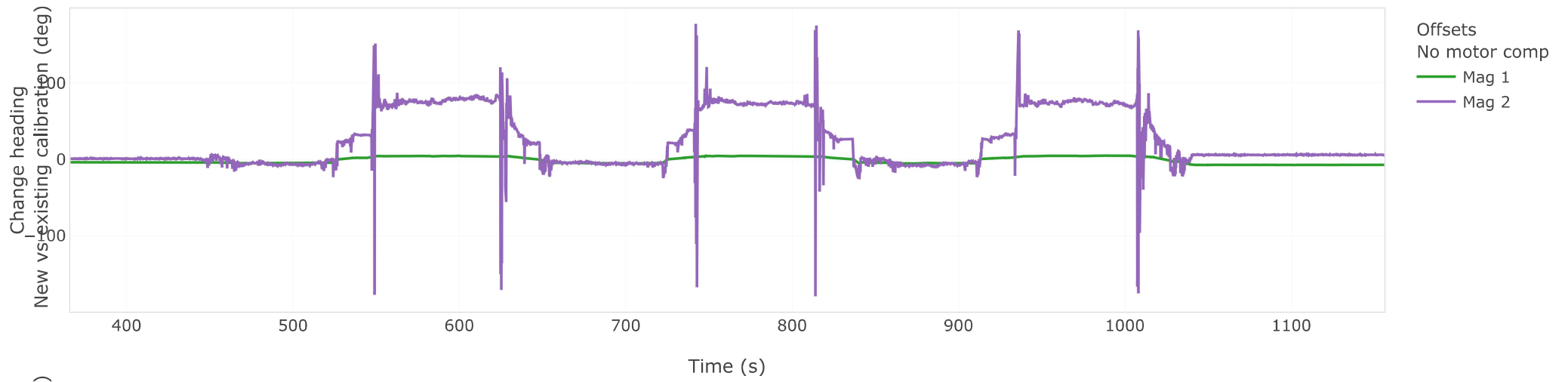
Compass 3

Not found

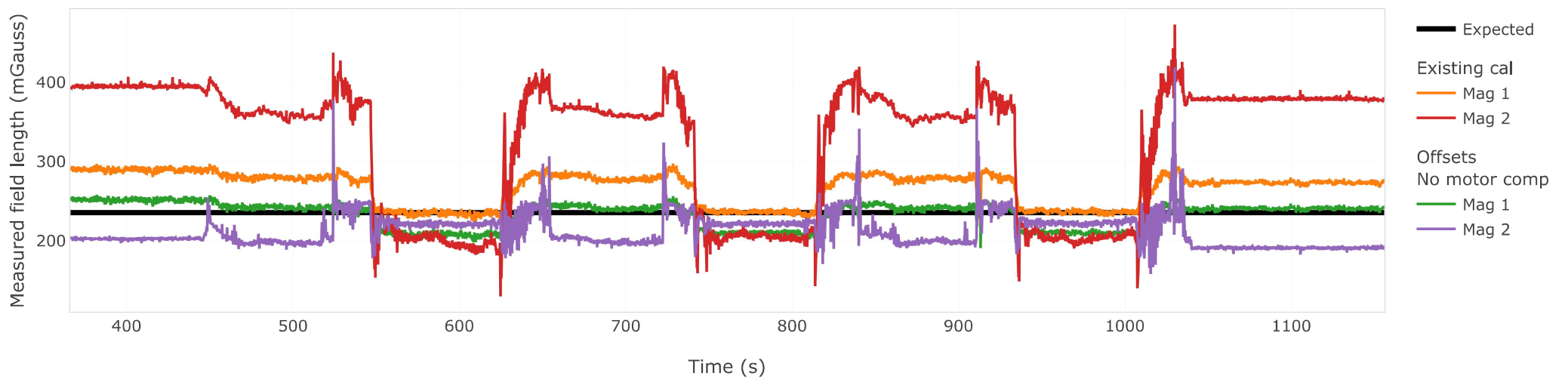




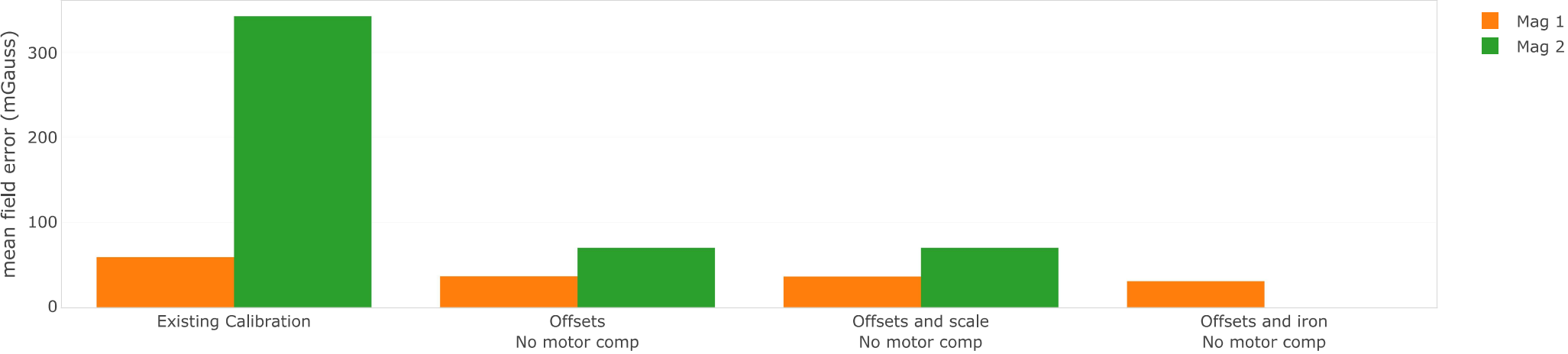
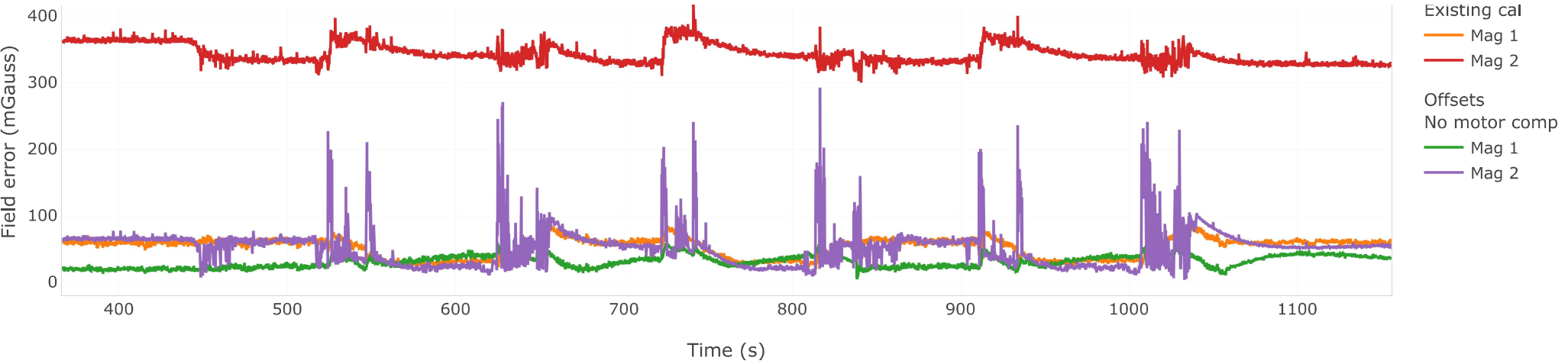
Yaw change (?)



Field length (?)



Calibration error (?)



Compass 1 parameters ?

Offsets, No motor comp

COMPASS_ORIENT

0:None

COMPASS_OFS_X

-93.403429666

mGauss

COMPASS_OFS_Y

20.998156071

mGauss

COMPASS_OFS_Z

13.362241846

mGauss

COMPASS_DIA_X

1

COMPASS_DIA_Y

1

COMPASS_DIA_Z

1

COMPASS_ODI_X

0

COMPASS_ODI_Y

0

COMPASS_ODI_Z

0

COMPASS_MOT_X

0

mGauss/A

COMPASS_MOT_Y

0

mGauss/A

COMPASS_MOT_Z

0

mGauss/A

COMPASS_SCALE

1

Compass 2 parameters ?

Offsets, No motor comp

COMPASS_ORIENT2

0:None

COMPASS_OFS2_X

-235.46027956

mGauss

COMPASS_OFS2_Y

-29.639704591

mGauss

COMPASS_OFS2_Z

-322.57351486

mGauss

COMPASS_DIA2_X

1

COMPASS_DIA2_Y

1

COMPASS_DIA2_Z

1

COMPASS_ODI2_X

0

COMPASS_ODI2_Y

0

COMPASS_ODI2_Z

0

COMPASS_MOT2_X

0

mGauss/A

COMPASS_MOT2_Y

0

mGauss/A

COMPASS_MOT2_Z

0

mGauss/A

COMPASS_SCALE2

1

Compass 3 parameters ?

Not found

COMPASS_ORIENT3

COMPASS_OFS3_X

mGauss

COMPASS_OFS3_Y

mGauss

COMPASS_OFS3_Z

mGauss

COMPASS_DIA3_X

COMPASS_DIA3_Y

COMPASS_DIA3_Z

COMPASS_ODI3_X

COMPASS_ODI3_Y

COMPASS_ODI3_Z

COMPASS_MOT3_X

mGauss/A

COMPASS_MOT3_Y

mGauss/A

COMPASS_MOT3_Z

mGauss/A

COMPASS_SCALE3

Save Parameters